

Sun Ju (Julie) Lee

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Updated 11/7/24

EDUCATION

Georgia Institute of Technology, Atlanta, GA 2019 – Present
H. Milton Stewart School of Industrial and Systems Engineering
Ph.D., Operations Research (Expected 2025)
Advisor: Gian-Gabriel Garcia
Thesis Committee: Lauren Steimle, Weijun Xie, Nicoleta Serban, Sheree Boulet
M.S., Operations Research 2022

Dartmouth College, Hanover, NH 2018
B.A., Engineering Sciences
B.E., Mechanical Engineering

PUBLICATIONS

Published and Accepted Papers

- [P2] **S. J. Lee**, X. Gong, and G.-G. P. Garcia (2024). Modified Monotone Policy Iteration for Interpretable Policies in Markov Decision Processes and the Impact of State Ordering Rules. *Annals of Operations Research*.
- [P1] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet (2023). Interpretable Machine Learning to Predict Adverse Perinatal Outcomes: Examining Marginal Predictive Value of Risk Factors During Pregnancy. *American Journal of Obstetrics & Gynecology MFM*, 101096.

Book Chapters

- [B1] **S. J. Lee**, H. S. Pandey, and G.-G. P. Garcia (2023). Designing Interpretable Machine Learning Models Using Mixed Integer Programming. In: Pardalos, P.M., Prokopyev, O.A. (eds) *Encyclopedia of Optimization*. Springer, Cham.

Working Papers

- [W4] Y. Yang, T. Liu, **S. J. Lee**, C.-Y. Liao, H. Shao, F. Pasquel, M. B. Weber, E. Keyvanshokoo, G.-G. P. Garcia (2024). Survival Modeling for CVD Risk Estimation Among a Diverse Cohort with Type-2 Diabetes.
- [W3] **S. J. Lee** and G.-G. P. Garcia (2024). A Data-driven Optimization Approach to Designing Parsimonious, Interpretable, and Equitable Treatment Guidelines.
- [W2] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet (2024). Composite Clustered Multi-Task Learning for Prediction of Adverse Pregnancy Outcomes.
- [W1] **S. J. Lee** and G.-G. P. Garcia (2024). A Tolerance-based Approach to Lexicographic Multi-Objective Markov Decision Processes.

PRESENTATIONS

Invited Conference and Workshop Presentations (Presenter)

- [C8] Y. Yang, T. Liu, **S. J. Lee**, C.-Y. Liao, H. Shao, F. Pasquel, M. B. Weber, E. Keyvanshokoo, G.-G. P. Garcia. Development and Fairness Evaluation of CVD Risk Prediction Models for Patients with Type 2 Diabetes. *Society for Medical Decision Making Annual Meeting*, Boston, MA, October 2024.
- [C7] **S. J. Lee** and G.-G. P. Garcia. A Data-driven Optimization Approach to Designing Parsimonious Treatment Guidelines. *INFORMS Annual Meeting (Bonder Scholars Session)*, Seattle, WA, October 2024.
- [C6] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet. Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes. *INFORMS Annual Meeting*, Seattle, WA, October 2024.
- [C5] Y. Yang, T. Liu, **S. J. Lee**, C.-Y. Liao, H. Shao, F. Pasquel, M. B. Weber, E. Keyvanshokoo, G.-G. P. Garcia. Survival Modeling for CVD Risk Estimation Among a Diverse Cohort with Type-2 Diabetes. *AI for Health Equity Symposium AIM-AHEAD Annual Meeting*, Atlanta, GA, August 2024.
- [C4] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet. Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes. *INFORMS Annual Meeting*, Phoenix, AZ, October 2023.
- [C3] **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet. Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes. *INFORMS Healthcare Conference*, Toronto, ON, July 2023.
- [C2] **S. J. Lee** and G.-G. P. Garcia. A Tolerance-based Approach to Lexicographic Multi-Objective Markov Decision Processes. *Society for Medical Decision Making Annual Meeting*, Seattle, WA, October 2022.
- [C1] **S. J. Lee** and G.-G. P. Garcia. A Tolerance-based Approach to Lexicographic Multi-Objective Markov Decision Processes. *INFORMS Annual Meeting*, Indianapolis, IN, October 2022.

Invited Seminar Presentations

1. Interpretable Machine Learning for Adverse Pregnancy Outcomes. Graduate Seminar Series, School of Industrial and Systems Engineering, University of Oklahoma, September 2023.

Other Presentations

3. **S. J. Lee** and G.-G. P. Garcia. A Data-driven Optimization Approach to Designing Parsimonious Treatment Guidelines. *INFORMS Annual Meeting Minority Issues Forum Poster Competition*, Seattle, WA, October 2024.
2. **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet. Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes. *The Next Generation 2023 Forum*, Atlanta, GA, October 2023.
1. **S. J. Lee**, G.-G. P. Garcia, K.H. Stanhope, M. H. Platner, and S. L. Boulet. Composite Clustered Multi-task Learning for Prediction of Adverse Pregnancy Outcomes. *INFORMS Annual Meeting Minority Issues Forum Poster Competition*, Phoenix, AZ, October 2023.

HONORS AND AWARDS

George Family Fellow

2024-2025

H. Milton Stewart School of Industrial & Systems Engineering, Georgia Institute of Technology

Finalist , Lee B. Lusted Prize in Quantitative Methods & Theoretical Developments <i>Society of Medical Decision-Making</i>	2022
President's Fellow , <i>Georgia Institute of Technology</i>	2019 – 2023
Virtual Annual Meeting & Membership Scholarship <i>Society of Medical Decision-Making</i>	2021
Thayer Scholar , <i>Dartmouth College</i>	2014 – 2018

STUDENTS MENTORED

Nathan Grodzinsky BS IE '24	2023 – 2024
Xingyu Gong BS IE '24	2022 – 2024

TEACHING EXPERIENCE

Teaching Assistant <i>Georgia Institute of Technology</i> , Atlanta, GA	2019 – 2021
– ISyE 3133 Engineering Optimization (undergraduate)	
– ISyE 6669 Deterministic Optimization (graduate)	
– ISyE 6661 Linear Optimization (graduate)	
– ISyE 4134/8813 Constraint Programming (undergraduate/graduate)	
<i>Dartmouth College</i> , Hanover, NH	2016 – 2017
– ENGS 76 Machine Engineering	
– ENGS 2 Integrated Design: Engineering, Architecture, and Building Technology	
– ENGS 25 Introduction to Thermodynamics	
Graduate Tutor <i>Georgia Institute of Technology</i> , Atlanta, GA	2020
– ISyE 2027 Probability with Applications	

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer <i>Epic Systems Corporation</i> , Verona, WI	2018 – 2019
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LEADERSHIP AND SERVICE ACTIVITIES

Volunteer <i>iExperience Summer Camp</i> , <i>Georgia Tech</i>	2024
Diversity, Equity, and Inclusion Committee Student Member <i>H. Milton Stewart School of Industrial and Systems Engineering</i> , <i>Georgia Tech</i>	2023 – Present
Graduate Research Mentor <i>Summer Undergraduate Research in Engineering/Sciences Program</i> , <i>Georgia Tech</i>	2022
Teaching Assistant <i>Seth Bonder Camp in Computational and Data Science for Engineering</i> , <i>Georgia Tech</i>	2020